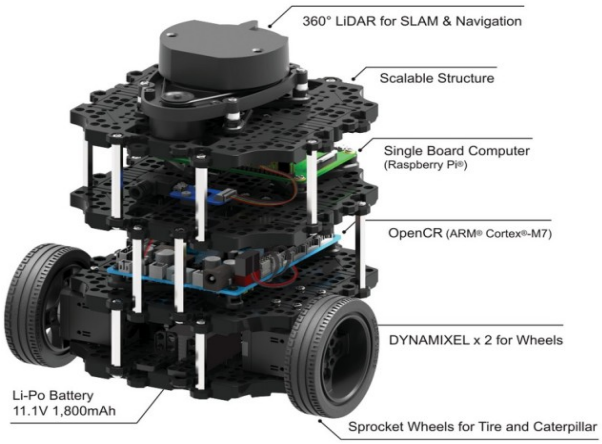
Indoor real-time navigation for robot vehicles

Final year project for the Embedded Systems Minor

The goal of the project is to design a real-time navigation system for a robot vehicle and to setup a complete demonstrator based on an existing robot (Turlebot 3 Burger, figure 1) operating under the ROS2 framework (Robot Operating System).



The first step is to implement the navigation system, to simplify, on a laptop running Ubuntu / ROS2 as a first instance (it will be ported to the main robot embedded computer in a second step). To this purpose, an acquisition device based on a portable autonomous laser scanner (Lidar) will be used to generate a map of the indoor navigation environment (Polytech buildings). A typical use case to build a map is the following: we want to be able to move the laser scanner around and collect data with it all while watching the map get created/updated in real time on a remote base station.

Figure 1. Turlebot 3 Burger.

The acquisition device is based on a Lidar sensor controlled by a mobile computer (laptop or embedded processor on robot) that can both communicate through a serial / USB connection. The Lidar data is streamed to a remote PC (workstation / laptop) running SLAM and visualization softwares. The acquisition device and the remote PC are connected through the Polytech Wifi network and follow the ROS2 communication model (topics).
The SLAM algorithm can be based on Cartographer or Gmapping, and a 3D visualization tool called RVIZ, both integrated in ROS2. The remote PC acts therefore as a base station for the system, remotely collecting Lidar samples and building the indoor navigation map in real time.

The recommended starting point for all developments is based on the offical documentation, available here :
https://docs.robotis.com/docs/systems/turtlebot3/overview/

The project is made up of three main parts that can be further decomposed into the following milestones :

1. Use Lidar and create a map with PC (workstation / laptop)
1.1 ROS2 setup and installation (Ubuntu 22.04 or 24.04 recommended)
1.2 ROS2 / RP Lidar demo
1.3 RP Lidar integration with map display (Cartographer or Gmapping) and Lidar demo from a laser scan bag file (workstation / laptop)
1.4. Build a first map of Polytech buildings

2. SLAM (Rasberry Pi 4)
2.1 Implement Gazebo simulation of Turtlebot3
2.2 Setup Ubuntu / ROS2 on microSD card for Rasberry Pi 4
2.3 Setup a Wifi connection between Rasberry and remote PC (workstation / laptop)

2.4 Implement SLAM with navigation display on remote PC
2.5 Setup and test Turtlebot3 teleoperation from remote PC
2.6 Create and save a complete map (and associated data) of Polytech buildings that will be further used for next Autonomous Navigation demo

3. Autonomous Navigation demo
3.1 Implement Turtlebot navigation from remote PC and using Polytech maps
3.2 Final demo showing Real Time autonomous navigation in Polytech buildings, with video
3.3 Final reporting, video demonstrators and GitLab

Organization, attendance and evaluation
1) The work is organized in a two-person team, each team works independently. Excessive similarity between two projects will suffer penalties in the final evaluation. Great work autonomy is an essential aspect of the project: each team must develop its own solution by himself, with minimum technical help from a professor. Project reporting will take place at each supervised session, essentially to check and assess the progress made on the project.

2) Final evaluation is based on the global project advancement (with respect to previous milestones), final completion, validation, quality of final solution/code/demonstration, and final report (all due to January 25, 2027). Final reporting will be limited to 5 pages maximum and structured as follows: Introduction / Specification / System overview, High-Level development / Hardware design / Software design, Tests and results, Conclusion (objective critique, possible improvements), References (articles, links, web), Appendices (code).

3) Quality of your code will be taken into account in the final evaluation. Your code must be accessible on Polytech Github server (https://gitlab.polytech.unice.fr/) and clearly structured into directories and sub directories. Special attention will be paid to the naming of files, functions, variables and proving meaningful comments. A main README.md must be present and explain the structure of your code and how to run the full demo. It must also contain a copy of your final pdf report. See for example:
https://gitlab.polytech.unice.fr/bilavarn/SLAM_Lidar_Lite_V3_Cartographer

4) Presence to supervised AND unsupervised project sessions are mandatory to ensure global project advances. Absence to supervised or unsupervised sessions will suffer penalties in the final evaluation.

5) To ensure a fair repartition of hardware between the different groups, robots are affected as follow :

Turtlebot #1 : G1_1 G1_2 G1_3 G1_4
Turtlebot #2 : G1_5 G1_6 G1_7 G1_8
Turtlebot #3 : G2_1 G2_2 G2_3 G2_4
Turtlebot #4 : G2_5 G2_6 G2_7 G2_8

The list of groups will be defined at the start of the project.

Any malfunctions and problem with the hardware will incur penalties in the final evaluation.